Lecture 1: Introduction

Statistical Foundations of Machine Learning

Aug 11, 2026

Course logistics

- *Instructors*: Nikhil Karamchandani (nikhilk@ee.iitb.ac.in) and D. Manjunath (dmanju@ee.iitb.ac.in)

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  1. Homeworks: 15

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Data Science: Probability and Statistics

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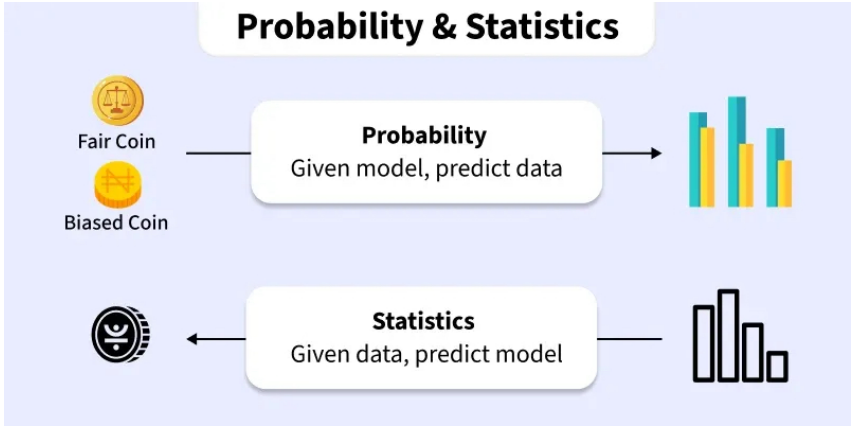


Illustration adapted from GeeksforGeeks:

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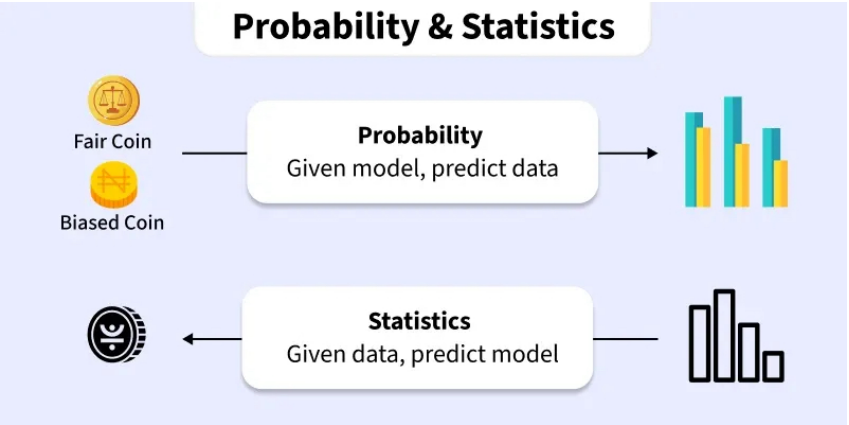


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- **Probability (Forward engine):**

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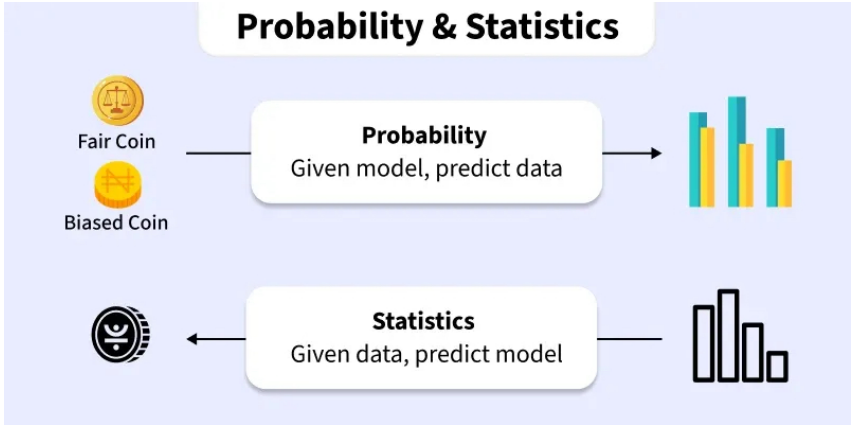


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Observation: Most days are stable; Day 5 is a massive outlier.

Descriptive Stats: Mean vs. Median

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Sorted: $[350, 380, 390, 400, \mathbf{410}, \mathbf{420}, 440, 460, 500, 8500]$

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- **Insight:** The Mean is inflated by the outlier. The Median better represents the "true signal" of daily spend.

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Verdict: The Mean is physically pulled by the magnitude of rare events. It fails to represent the majority of days.

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Verdict: Median follows frequency, not magnitude. Better choice as the behavioral baseline for this user.

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- Using our empirical Median value ($\tilde{x} = 415$):

$$\lambda = \frac{0.693}{415} \approx \mathbf{0.00167}$$

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Decision: Risk (3.5%) is below our 5% threshold. **Loan Approved.**

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- **Question:** Are the observed differences in $\hat{p}_A$ and $\hat{p}_B$ real, or due to random sampling noise?

Probabilistic Model and CLT

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- By the Central Limit Theorem (for large n):

$$\hat{p} \approx \mathcal{N}\left(p, \frac{p(1-p)}{n}\right)$$

From Gaussian Fluctuations to Confidence

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- Interpretation:

$$\hat{p} \text{ typically lies within } \pm 1.96\sqrt{\frac{p(1-p)}{n}} \text{ of } p$$

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Conflict: Intervals overlap. The observed lift may be due to randomness.

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Scenario: Increase sample size to **5,000** users per group.

$$SE_B = \sqrt{\frac{0.34 \cdot 0.66}{5000}} \approx \mathbf{0.0067} \;\Rightarrow\; CI_B = [\mathbf{32.7}\%, \mathbf{35.3}\%]$$

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Decision: No overlap remains. The observed 4% lift is unlikely to be noise. **Deploy Gold.**

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Goal: Distinguish meaningful signal from random noise.

Comparative Correlation Table

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A	15	600	50	-11	-550	6,050	0	0
B	21	800	85	-5	-350	1,750	35	-12,250
C	25	1100	10	-1	-50	50	-40	2,000
D	31	1250	45	5	100	500	-5	-500
E	38	2000	60	12	850	10,200	10	8,500
Sum						**18,550**		**-2,250**

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A	15	600	50	-11	-550	6,050	0	0
B	21	800	85	-5	-350	1,750	35	-12,250
C	25	1100	10	-1	-50	50	-40	2,000
D	31	1250	45	5	100	500	-5	-500
E	38	2000	60	12	850	10,200	10	8,500
Sum						**18,550**		**-2,250**

Next: Normalize these sums to obtain correlation coefficients.

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Conclusion: Session Time is a useful leading indicator; Screen Brightness is not.

Conclusion: Statistics and Probability in Data Science

Core Theme: Data science is about making decisions under uncertainty.

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We will study these notions formally in this course!!

Tentative lecture plan - first half

Lec1: Introduction
Lec 2: Probability axioms, conditional probability and independence,
Lec 3: Counting and sampling
Lec 4: Random Variables, CDF, Joint CDF, Independence
Lec 5: Discrete RVs (PMF), Joint / Conditional PMF
Lec 6: Continuous RVs (PDF), Joint / Conditional PDF
Lec 7: Expectation
Lec 8: Variance, Covariance and Correlation
Lec 9: Moment generating functions
Lec 10: Tail Inequalities and applications
Lec 11: Law of Large Numbers and Central Limit Theorem
Lec 12: Conditional expectation and MMSE

References

Books:

- **A first course in probability, S. Ross**
- Probability, Random Variables and Stochastic Processes, Papoulis and Pillai
- Probability with Engineering Applications, Bruce Hajek, available at https://courses.grainger.illinois.edu/ece313/fa2020/probabilityAug21.pdf
- Probability and Random Processes, Grimett and Stirzaker
- One Thousand Exercises in Probability by Grimmett and Stirzaker.

Online resources:

- https://ocw.mit.edu/resources/res-6-012-introduction-to-probability-spring-2018/index.htm
- https://nptel.ac.in/syllabus/108106083/ (Probability foundations, Krishna Jagannathan, IITM)